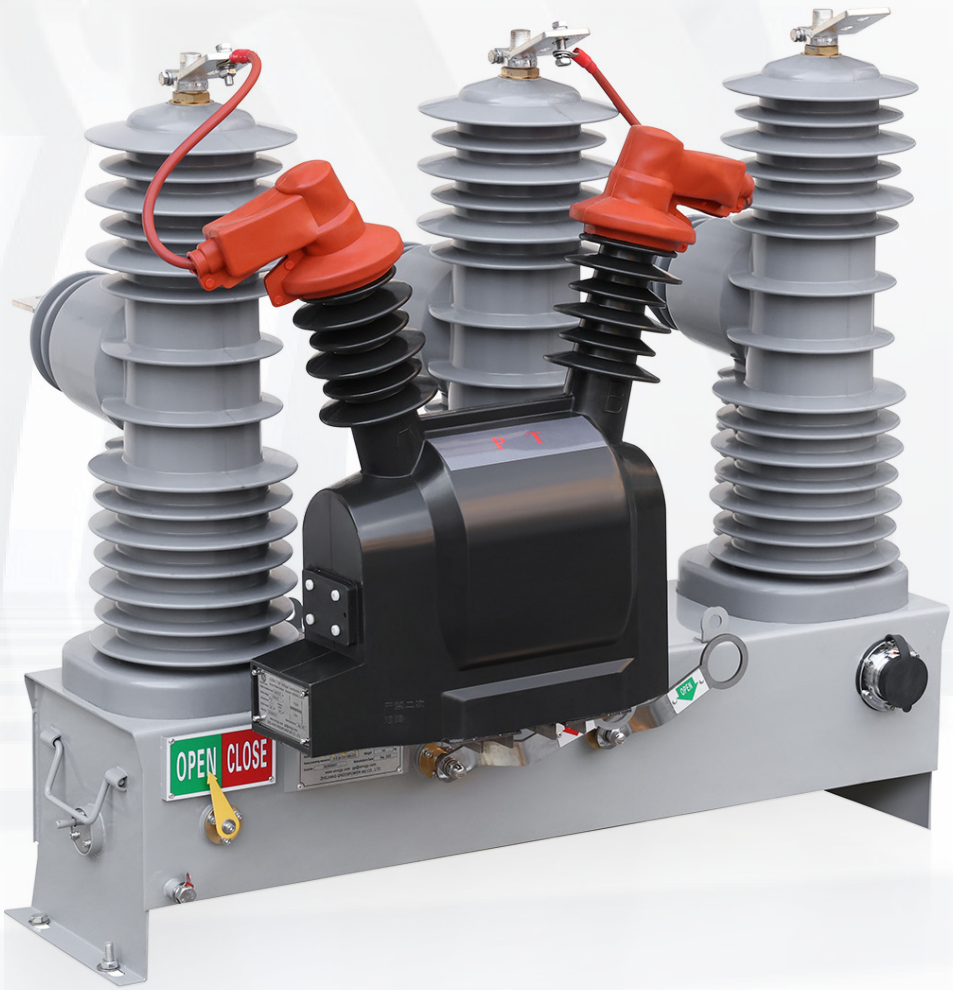
Installation and Operation Manual

GreenPower

ZW32-17.5(24)

Outdoor High Voltage Vacuum Circuit Breaker (Recloser)





About Us»

GreenPower is a manufacturer of medium- and low-voltage switchgear and components, with in-house engineering, manufacturing and testing capabilities. Our manufacturing entity, Zhejiang GreenPower Electric Co., Ltd., was established in 1992. International business and export operations are handled by Zhejiang GreenPower I&E Co., Ltd.

Our product portfolio covers medium-voltage switchgear from 12 kV to 40.5 kV, including air-insulated metal-clad switchgear (AIS), cubicle-type gas-insulated switchgear (C-GIS), ring main units (RMU), vacuum circuit breakers, and a broad range of switchgear components and accessories. We also manufacture low-voltage withdrawable switchgear systems with ratings up to 6300 A.

Engineering and manufacturing are central to what we do. Our production and testing capabilities include automated sheet-metal processing, CNC laser cutting, 3D five-axis laser welding for stainless-steel gas tanks, automatic helium leak detection, and a fully shielded partial-discharge test room. These capabilities support consistent manufacturing quality from enclosure fabrication through assembly and final testing.

For customers moving toward lower-impact power distribution systems, GreenPower offers SF6-free solutions including dry-air insulated RMUs and nitrogen-insulated C-GIS, alongside conventional gas-insulated solutions where required. These technologies help customers respond to increasingly stringent environmental requirements for medium-voltage switchgear.

We also support project-specific engineering based on customer drawings and specifications, including non-standard dimensions, special voltage and frequency requirements, high-altitude applications, customized component selection and other application-specific configurations.

Our goal is simple: to be the switchgear manufacturer our customers return to — because the equipment is properly engineered, reliably manufactured and built for long-term service.

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01 Product Overview

1.1 The main purpose

The ZW32-17.5(24) outdoor pole-mounted high-voltage vacuum circuit breaker (hereinafter referred to as the circuit breaker) is an outdoor power distribution equipment with a rated voltage of 17.5/24 kV, three-phase AC, 50 Hz. It is primarily used to disconnect and connect load currents, overload currents, and short-circuit currents in distribution networks. It is suitable for protection and control in substations and industrial and mining enterprise distribution systems. It is particularly well-suited for rural power grids and locations with frequent operation, and is particularly well-suited for urban and rural power grid upgrades.

This installation and operating manual specifies the main technical parameters, product structure, and principles and methods of operation, installation, use and maintenance of the circuit breaker.

1.2 Execution standard

IEC62271-100 "High-voltage AC Circuit Breakers"

GB1984 "High-voltage AC Circuit Breakers"

GB/T11022 "Common Technical Requirements for High-voltage Switchgear and Controlgear Standards"

GB311.1 "Insulation Coordination for High-voltage Transmission Transformers"

GB3309 "Mechanical Tests for High-voltage Switchgear at Normal Temperatures"

DL403 "Technical Conditions for Ordering 10-35kV High-voltage Vacuum Circuit Breakers"

JB/DQ2080 "Rainproof Test Method for High-voltage Switchgear"

GB2706 "Dynamic Thermal Stability Test Method for AC High-voltage Electrical Apparatus"

GB763 "Heat Generation During Long-term Operation of AC High-voltage Electrical Apparatus"

DL/T721 "General Technical Conditions for Distribution Automation Terminal Equipment"

1.3 Environmental conditions

- Altitude: 0~2000 meters
- Ambient air temperature: -40℃ ~ +45℃
- Wind speed: ≤ 40 m/s
- Pollution level: Class IV
- Installation location: Free of flammable, explosive, or chemically corrosive materials
- Earthquake intensity: ≤ 8 degree

Note: Please contact us if the environment exceeds the above specifications.

1.4 Product model and meaning

ZW 32 - 17.5/24 - T □ - □

- Rated short-circuit breaking current (kA)
- Rated current (A)
- Operating mechanism: T: spring, M: permanent magnet (electric and manual)
- Rated voltage (kV)
- Design number
- Outdoor vacuum circuit breaker

1.5 Main features

- The three-phase, pillar-type circuit breaker is compact, lightweight, clean, and easy to operate and maintain.
- The circuit breaker can be opened and closed manually, electrically, or remotely. It can be used with a controller to achieve power distribution automation or as a recloser.

- The spring-operated mechanism is equipped with an FDK-10 composite controller. This controller provides combined overcurrent protection for the power grid during inrush current and short circuit conditions. The device's protection modes include: inrush current delay + overcurrent fault delay + short-circuit fault definite time instantaneous action.

Advantages of this device:

- The inrush current time is adjustable to 100ms, 150ms, 200ms, 300ms, 500ms, 1000ms, and 1500ms.
- While ensuring the elimination of false tripping caused by closing inrush current and reliably eliminating overcurrent trip coil chatter, this circuit breaker not only eliminates false tripping caused by transient overcurrent faults but also ensures rapid interruption of short-circuit faults, significantly improving power supply reliability.
  - ★ Fully outdoor design, with strong anti-interference, corrosion, and condensation resistance
  - ★ Compact size and easy installation
  - ★ Low cost and strong adaptability
  - ★ Eliminates trip coil chatter
  - ★ Suitable for three-phase protection
- The circuit breaker can be equipped with two-phase or three-phase CTs, providing information analysis for the overcurrent automatic trip protection and intelligent controller.

02 Technical Parameters

2.1 Main technical parameters table

No.	Item		Unit	Data
1	Rated voltage		kV	17.5, 24
2	Rated current		A	630, 1250
3	Rated frequency		Hz	50
4	Rated peak withstand current		kA	50, 63
5	Rated short-time withstand current (4s)		kA	20, 25
6	Rated short-circuit breaking current		kA	20, 25
7	Rated short-circuit making current (peak value)		kA	50, 63
8	Mechanical life		times	Spring mechanism: 10000 Permanent magnet mechanism: 30000
9	Power frequency withstand voltage (1 min)	To ground, phase-to-phase (wet)	kV	50
		To ground, phase-to-phase/break (dry)	kV	65, 79
10	Rated short-circuit breaking current interruption times		times	30
11	Lightning impulse withstand voltage (peak value)		kV	125
12	Rated operating sequence			O - 0.3s - CO - 180s - CO
13	Secondary circuit power frequency withstand voltage (1 min)		V	2000
14	Weight		kg	195

2.2 Mechanical properties table

No.	Item	Unit	Data
1	Contact gap	mm	12 ± 1
2	Contact over-travel	mm	3 ± 1
3	Opening speed	m/s	1.4 ± 0.35
4	Closing speed	m/s	0.7 ± 0.25
5	Contact closing bounce time	ms	≤ 2
6	Three-phase asynchronism	ms	≤ 2
7	Inter-phase center distance	mm	380 ± 2
8	Opening time	ms	20~50
9	Closing time	ms	25~80
10	Resistance of conductive path (per phase)	uΩ	≤ 200

2.3 Main technical parameters of external voltage transformer

No.	Item	Unit	Data
1	Model		JDZW
2	Rated voltage	kV	17.5, 24
3	Rated frequency	Hz	50
4	Accuracy		Class 1
5	Transformation Ratio	kV/ kV	17.5/0.22
6	Limit Output	VA	500
7	Weight	kg	25

2.4 Current transformer

Current transformers with ratios of 50/5, 75/5, 100/5, 150/5, 200/5, 300/5, 400/5, 500/5, 600/5, 750/5, 1000/5, and 1250/5 can be installed according to user requirements. Current transformers with multiple ratios, such as 100-200-300/5, 200-400-600/5, and 600-800-1000/5, can also be installed based on actual needs.

The accuracy grades are 0.2S, 0.2, 0.5S, 0.5, and 10P, which can be combined into 0.2S/10P, 0.2/10P, 0.5/10P, 0.2S/0.5/10P, 0.2S/0.5/10P/10P, etc.

03 Product Structure and Working Principle

3.1 Product structure

The circuit breaker consists of an operating mechanism, conductive circuit, insulation system, seals, and housing. The overall structure is a three-phase pillar-type. The conductive circuit is formed by connecting the incoming and outgoing conductors, the moving and static end conductive plates, the conductive clamps, and the vacuum arc extinguishing chamber. External insulation is primarily achieved through high-voltage porcelain sleeves, which are resistant to contamination. Internal insulation is composite insulation.

3.2 Operation principle

The circuit breaker is operated via an operating mechanism. The mechanism and trip spring rotate the breaker's three-phase main shaft, simultaneously pulling the insulated operating lever and crank arm, which opens or closes the moving and static contacts of the vacuum interrupter, placing the circuit breaker in the open or closed state. After the circuit breaker is closed, pulling the manual trip ring or applying voltage to the trip coil causes the trip spring to drive the operating lever, opening the circuit breaker.

During operation, when the line current exceeds the set value, the overcurrent coil activates, tripping the circuit breaker and protecting the line and equipment. However, this protection tripping is limited by accuracy. For high-accuracy protection of the outgoing load, please configure a dedicated relay protection device or intelligent controller.

3.3 Product structure size

- **Vacuum circuit breaker**
  Dimensions: 1032mm(length) * 266mm(width) * 880mm(height).
- **Vacuum circuit breaker + Isolating switch**
  Dimensions: 1080mm (length) * 780mm (width) * 880mm (height).
  Our company can install isolating switches on both the incoming and outgoing lines according to different user requirements. A mechanical interlock is set between the isolating switch and the circuit breaker, and the circuit breaker can only be closed when the isolating switch is closed.
- **Vacuum circuit breaker + External dry PT**
  Our company can provide single or dual-side PTs to meet different user requirements. This allows the circuit breaker to have its own operating power supply, providing a highly reliable operating power source for the controller, solving the problem of external power supply for outdoor products. In conjunction with intelligent controllers, this allows for automated power distribution.
- **Vacuum circuit breaker + Intelligent controller**
  Reference dimensions of the intelligent controller are: 500mm*400mm*200mm.
  For the safety of operators and maintenance personnel, the intelligent controller is usually mounted separately on a pole below the circuit breaker.
- **Circuit breaker weight:** Approximately 190kg to 210kg.

04 Transportation, Commissioning and Installation

- The product must not be turned over or inverted during transportation, and anti-vibration measures must be taken. When lifting the circuit breaker, you must hook the two earrings on the box and lift it horizontally.
  After unpacking, check whether the outer insulation of the circuit breaker's insulating sleeve is broken, whether the box is deformed, whether the opening and closing instructions are complete, whether the product nameplate and certificate are consistent with the order requirements, and whether the packing list is consistent with the actual product.
- Before commissioning the circuit breaker, first inspect the product's porcelain sleeve for damage, any missing opening and closing indicators, and any missing energy storage indicators, as well as any deformation of the circuit breaker. Then, open the circuit breaker and place the cover on a shelf. Check the bottom for any loose metal objects, cracks in the arc extinguishing chamber, or loose fasteners. Then, perform a mechanical property check based on the data listed in Table 2. After inspection, replace the cover on the circuit breaker and secure it evenly with screws. Once sealed, proceed with installation. This product should be installed on a pole at least 4 meters tall.
- The circuit breaker can be mounted on a single pole or a double pole. The circuit breaker should be installed stably and securely on a dedicated steel frame.

05 Product Inspection and Maintenance

- The product is inspected every five years to check the wear of the moving parts and inject some lubricant into the moving parts. The product has a shelf life of 15 to 20 years.
- Users are not allowed to arbitrarily replace and use electrical components that are inconsistent with the original model and specifications.
- Operators should have a preliminary understanding of the performance of the mechanism and knowledge of installation, commissioning and maintenance, record any problems encountered during operation, and notify the manufacturer when necessary.

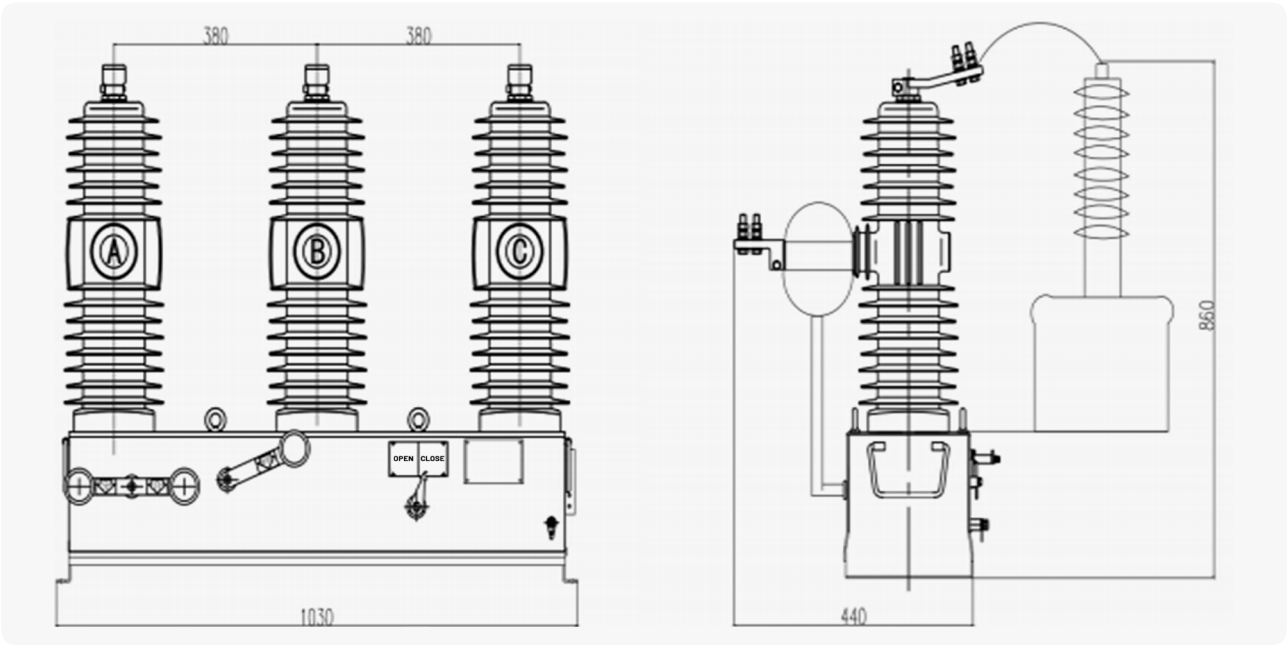
06 Random Files

- One product quality certificate.
- One installation and operation manual.
- One packing list.

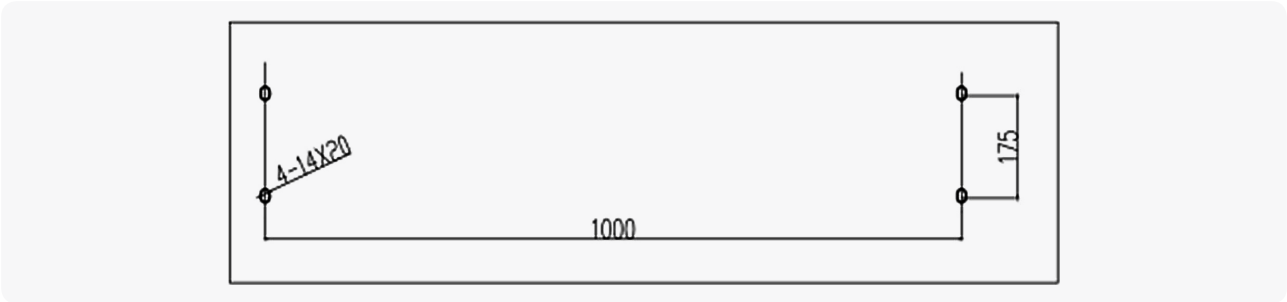
07 Ordering Instructions

When placing an order, please specify the product model, name, quantity, rated current, current ratio of the current transformer, operation method and application.

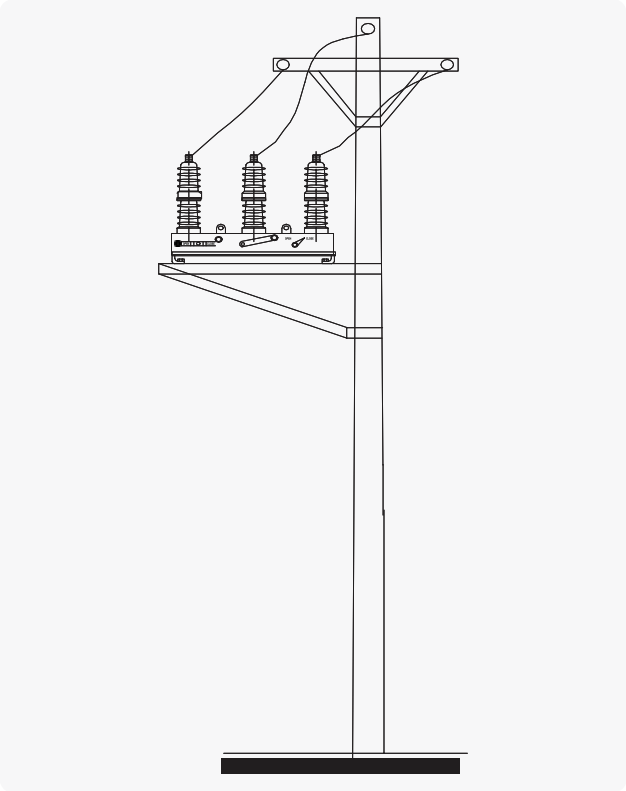
08 Outline Dimension and Installation Base Drawings



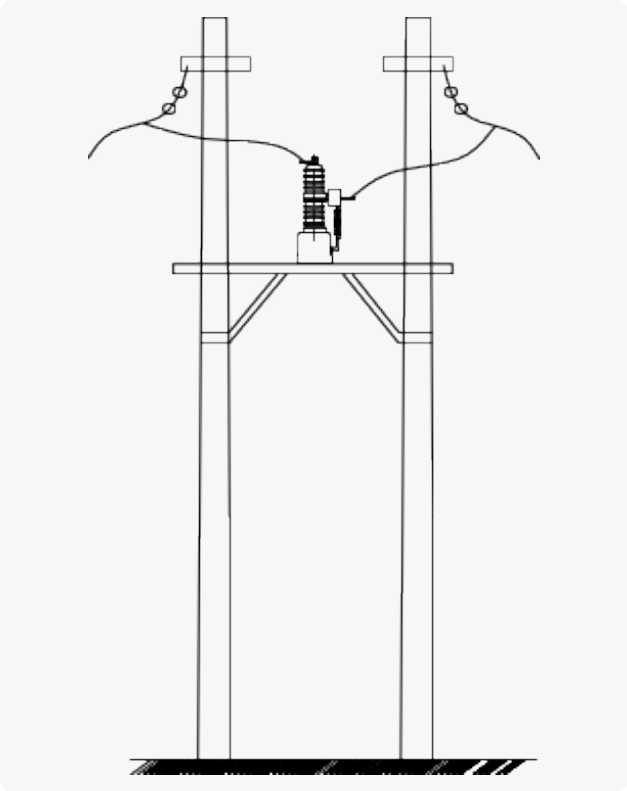
Outline dimension drawing



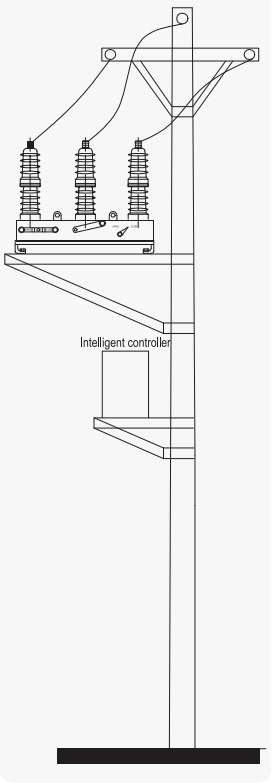
Standard product installation base drawing



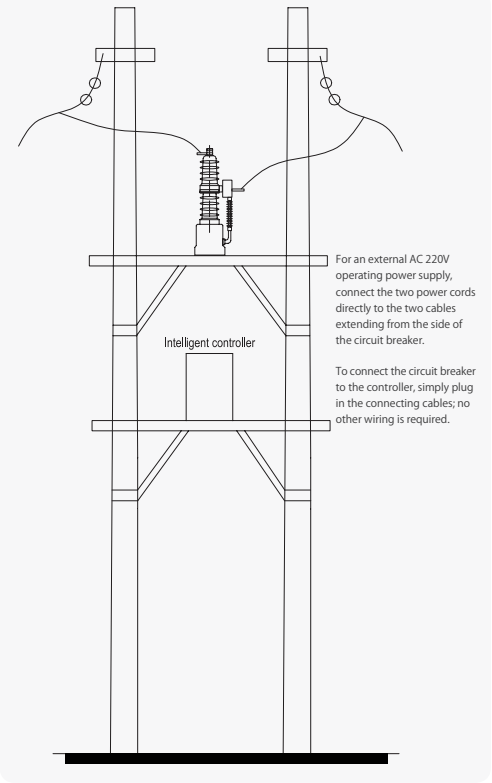
Single pole installation



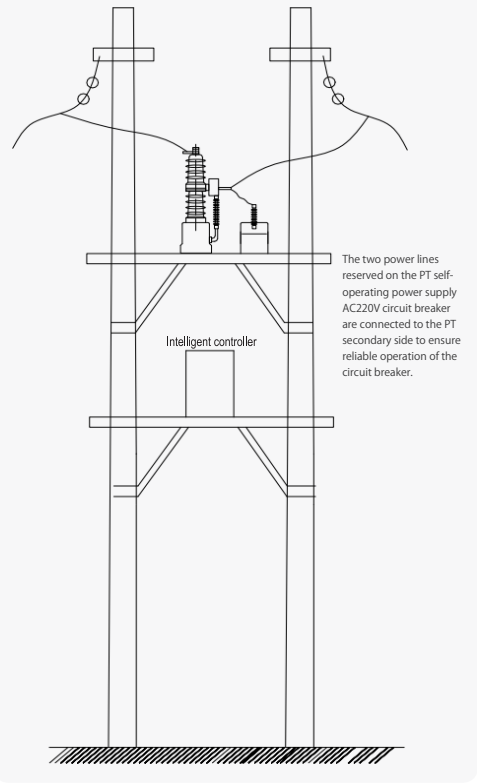
Double pole installation



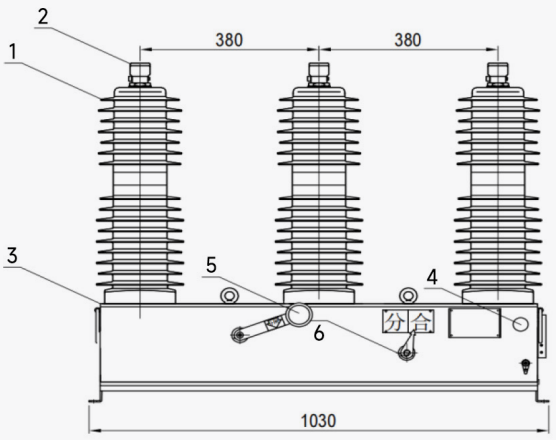
Single pole installation

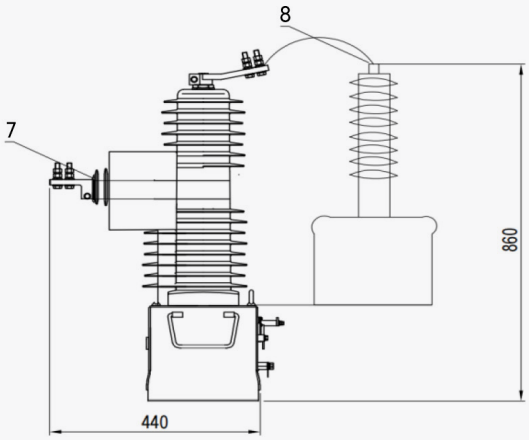


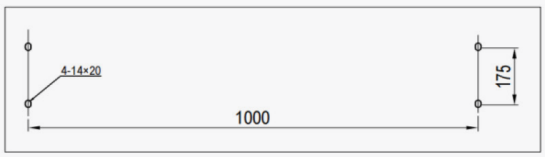
Double pole installation



Double pole installation







Technical Parameters:

Rated voltage: The rated voltage of the equipment is 17.5kV (suitable for use at a rated voltage of 24kV).

Rated current: 1250A

Rated breaking Current: 20kA

Mechanical life (Permanent magnet mechanism): 30000 times

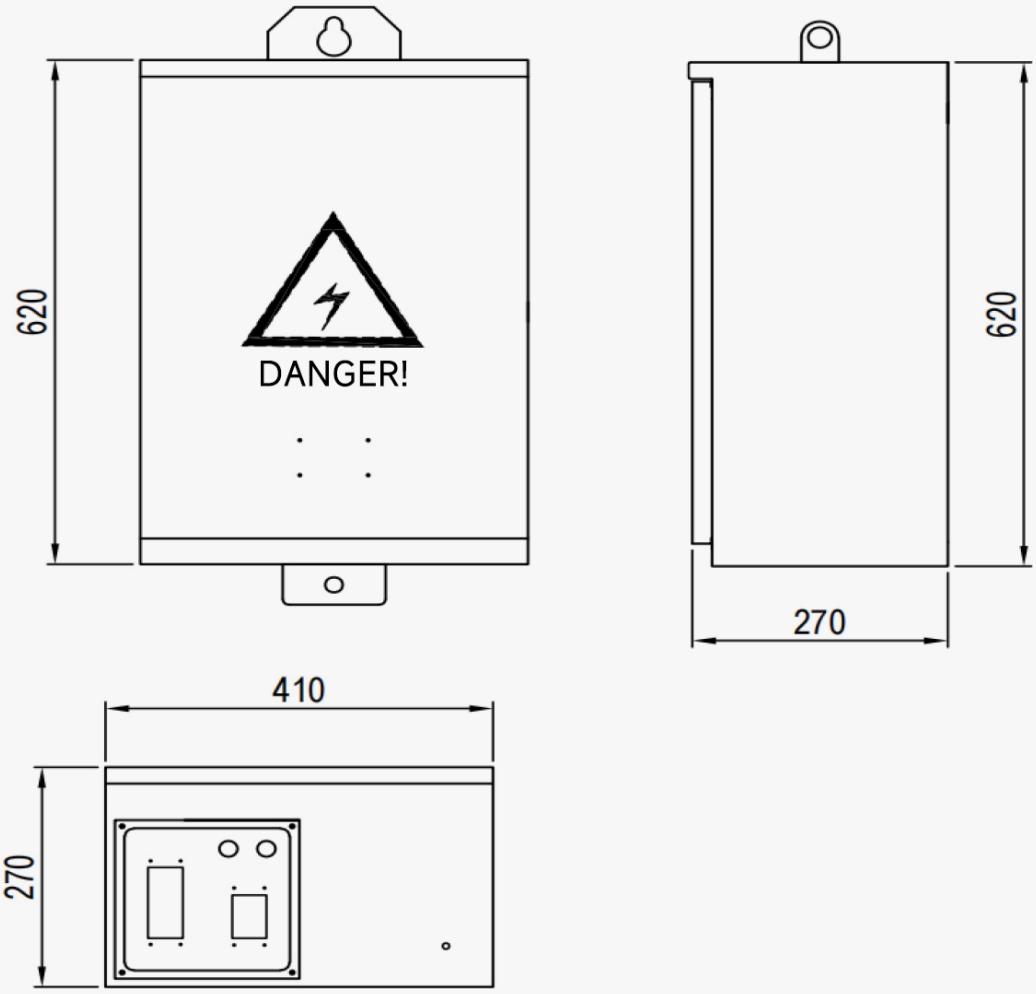
Power frequency withstand voltage: 60/65kV

Rated peak withstand current: 50kA

(Note: The product has full English marking.)

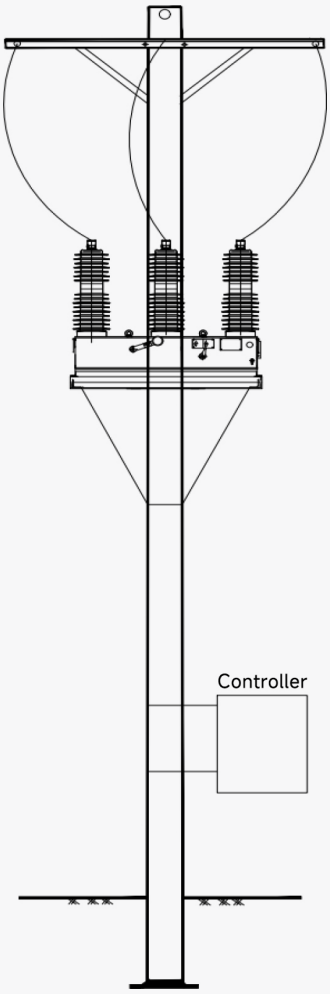
No.	Item	Qty.	Remark
1	ZW32-24kV (Special) Insulator bushing	3 pcs	
2	24kV Inlet and outlet cable connectors	6 pcs	
3	24kV Enclosure	1 set	
4	ZW32-24kV Secondary plugin	1 set	
5	ZW32-24kV Open handle	1 pcs	
6	ZW32-24kV Open/Close indicator	1 pcs	
7	LZZB-32 Current transformer (CT)	3 pcs	200-400-600/5
8	JDZ-24W 17.5/0.22 Voltage transformer (PT)	1 pcs	Power supply function, without voltage monitoring.
9	TD-24 Vacuum interrupter	3 pcs	Internal configuration
10	YTB Permanent magnet mechanism	1 set	Internal configuration

ZW32-17.5/1250-20 Reclosing circuit breaker



No.	Item	Qty.	Remark
1	620*410*270mm Enclosure	1 set	
2	8 meter cable (for controller)	2 pcs	
3	SAF5CE Core control unit	1 pcs	Internal configuration
4	JW300-24BC-220T-13X Charging module	1 pcs	Internal configuration
5	SR302 Communication module	1 pcs	Internal configuration
6	6-FM-7 7AH Lead-acid battery	1 pcs	Internal configuration

SAF5-CE Performance parameters	
Power supply	AC 220V ± 20%, dual power supply
Mechanism type	with permanent magnet mechanism
Number of mechanism operations	30,000 times
Capacitor capacity	47000uF/250V
Control power outage operation times	≥ 50
Battery capacity	24V/7Ah
Rated phase voltage	100V
Rated zero-sequence voltage	100V
Rated phase current	1A/5A, default 5A
Rated zero-sequence current	1A/5A, default 1A
Core unit operating voltage	DC 24V
Power consumption	≤ 20W
AC Voltage overload capacity	1.5 times the rated voltage, continuous
AC Current overload capacity	2 times the rated current: continuous 20 times the rated current: allow 1 second
Accuracy	Voltage error: ≤ 0.5% Current error: ≤ 0.5% Active power and reactive power error: ≤ 1% Power acquisition error: ≤ 1% Frequency error: ± 0.02Hz
SOE resolution	≤ 2 ms
Software filtering time	5~60000ms configurable
Shutter protection inherent action time	≤ 20 ms
Guard time accuracy	≤ 40 ms
Date accuracy	Error of no more than 1s per 24 hours
Protection level	IP55
Communication interface	RS232×2, RS485×1, RJ45 Ethernet access
Communication protocols	IEC61850/IEC60870-5-101/104, DNP3.0, Modbus-RTU
SCADA	Optional USB, Ethernet
SMC SMS function	Including SMS function
Protective devices	Cold load pickup (no dedicated ANSI code) Non-interruptible blocking (ANSI 60), inrush current blocking (ANSI 68), overcurrent protection (ANSI 50/51), overload protection (ANSI 49), zero-sequence protection (ANSI 50N/51N) Small ground fault protection (ANSI 64N) Overcurrent inverse time (ANSI 51V), zero current inverse time (ANSI 51N), overvoltage protection (ANSI 59), undervoltage protection (ANSI 27), frequency protection (ANSI 81), line interruption protection (ANSI 40), reclosing (ANSI 79) Synchrocheck protection (ANSI 25)



24kV Recloser installation diagram (Single pole installation)

E5.0105ME-202509

Zhejiang Greenpower Electric Co., Ltd
Zhejiang GreenPower I&E Co., Ltd
Professional Power Equipment Provider

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